ACTVET

Abu Dhabi Centre for Technical and Vocational Education and Training

مركز أبوظبي للتعليم والتدريب التقني والمهني

MOBILE ROBOTICS

Training Program

Skills Challenge

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Mobile Robotics Training Plan – Skills Challenge

Objective:

The objective of this training program in Mobile Robotics is to equip students with the necessary theoretical knowledge and practical skills to excel in the mobile robotics industry. The program aims to produce skilled professionals who can effectively design, prototype, program, and test modern mobile robotics systems while adhering to safety and environmental standards.

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Hardware - Mechanics

This module introduces the mechanical fundamentals essential for building and understanding mobile robots. It covers key mechanical components, structures, and motion systems, focusing on how robots move and interact with their environment.

Topic	Subject	Task
Introduction to mechanics	Basic mechanical concepts: force, torque, and friction in robotic mechanisms	Theorical Approach
	Mechanical constraints, tolerances, and clearances in robot assemblies	
	Balance, stability, and center of gravity in mobile robots	
Bearings, bushings, and axles	Role of bearings and bushings in reducing friction	Explore how rotating components support movement and reduce friction in mechanical systems.
	Differences between bearings and bushings	
	Axle design and load support in mobile robots	
	Alignment and mounting of rotating components	
Gears, pulleys, and linkages	Types of gear used in robotics (spur, bevel, worm)	Explore how mechanical transmission components transfer motion and modify speed, torque, and direction in robotic mechanisms.
	Gear ratios and torque-speed relationships	
	Pulley systems and belt transmissions	
Chassis design and construction	Types of wheels and its implementation in a robot frame: Omnidirectional, Mecanum, All-terrain Types of robot chassis (differential, omnidirectional)	Build small chassis structures with different levels of mechanical reinforcement and compare their rigidity and resistance to deformation.
	Structural rigidity and load distribution	
	Mounting components on the chassis: motors, sensors, batteries	
Object manipulation system (OMS)	Types of end-effectors (grippers, claws, suction)	Design and assemble a basic elevator and gripper mechanism and test its ability to grasp and manipulate different objects.
	Mechanical design of grippers and lifting	
	Degrees of freedom in manipulation mechanisms	
	Force and grip considerations for object handling	
Safety and Ergonomics	Safe handling of tools and mechanical components	Identify potential mechanical hazards and apply basic safety and ergonomic principles during robot assembly
	Identifying mechanical hazards in robotic systems	
	Ergonomic considerations when assembling robots	

Outcomes:

- Recognize and explain the purpose of wheels, gears, shafts, bearings, frames, and fasteners.
- Explain concepts such as torque, force, friction, leverage, and mechanical advantage.
- Use mechanical tools to construct stable and functional chassis and frames.
- Select materials and mechanical parts based on design needs (strength, weight, cost).
- Identify and resolve common issues like misalignment, instability, or excessive friction.

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- Use tools safely and maintain awareness of potential hazards during mechanical work.

Hardware - Electronics

This module introduces participants to the foundational concepts of electronics necessary for building and controlling mobile robots. It covers basic electronic components, circuit design, and safe handling practices.

Topic	Subject	Task
Introduction to Electronics	Basic electrical concepts: voltage, current, and resistance (Ohm's Law basics)	Use multimeter to measure voltage, resistance, current and continuity in electronic components
	Power supply types (batteries, voltage regulators)	
	Understanding polarity and basic safety	
Microcontroller Fundamentals	Introduction of the microcontroller devices	Follow a basic schematic diagram to connect the power supply to the microcontroller correctly
	Digital and analog inputs and outputs concepts	
	Pin configurations: digital, analog, PWM	
	Power management on microcontrollers	
Motors and Actuation	Types of motors used in robotics: DC, servo, stepper	Follow a basic schematic diagram to connect motors to motor drivers and microcontroller correctly
	Motor drivers and power control	
	PWM for direction and speed control	
Sensors and Signal Processing	Role of sensors in robotic perception	Follow a basic schematic diagram to connect sensors to the microcontroller correctly
	Overview of basic sensors: Ultrasonic, IR, Line Follower, Limit Switch, etc.	
	Analog and digital signals	
Troubleshooting and Debugging	Using multimeters to check continuity and voltage	Analyze a circuit with an intentional error and work together to identify and fix the problem
	Common faults: short circuit, open circuits, polarity reversal	

Outcomes:

- Assemble and test basic circuits using standard prototyping tools.
- Measure voltage, current, and resistance accurately with a multimeter.
- Understand different power supply options and how to safely connect them to circuits.
- Explain how microcontrollers act as the brain of a robot by reading sensor data and controlling outputs.
- Identify common safety risks and follow procedures to prevent short circuits, overheating, and damage.

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Software Environment

This module introduces participants to the software tools and environments used for programming mobile robots. It covers how to install, configure, and navigate essential development platforms such as IDEs (e.g., Arduino IDE, VS Code), compilers, and drivers.

Topic	Subject	Task
Software Setup	Software installation	Create a complete and functional project
	Setting up VS Code for the VMX	
	Creating a project	
	Adding Vendor Libraries	
	Control Station Console Installation	
WPILib and coding execution	Ussing Control Station Console	Test the project execution with the Control Station Console
	Adding MockDS to the project	Test the project execution with the MockDS

Outcomes:

- Explain how IDEs, compilers, and toolchains support the programming workflow.
- Start new projects, organize files, and save code versions properly.
- View live output from a program and understand how to use it for testing.
- Add third-party libraries and ensure hardware drivers are correctly installed.

Basic programming:

To equip participants with fundamental C++ programming skills essential for controlling mobile robots in competitive environments.

Topic	Subject	Task
Basics concepts of C++	Structure of a program in C++. Shuffleboard implementation	Write a basic C++ program that turns on the green LED and sends a value to Shuffleboard to indicate that the program has started.
	Libraries and implementation of delays in code execution.	Write a basic C++ program to control the speed of LED blinking.
	Variables, data types and constants	Create a bool variable to store the current LED state and one more to count the blink intervals, then display the values in Shuffleboard.
	Arithmetic, assignment, logical and comparison operators	1. Write a program that toggles the green LED based on a counter variable and use arithmetic and comparison operators to change the blink pattern after a certain number of iterations.
Control Structures	Conditional statements structures If – else and switch – case	

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		2. Program the green LED to turn on only when the Start button is pressed and turn off when it is released, using a simple condition to control LED state. 3. Implement multiple system states using if–else or switch–case: • Start pressed → green LED ON • Stop pressed → green LED OFF • Emergency pressed → red LED ON
	Iteration structures For, while and do-while	Program the VMX controller to count how many times the Stop button is pressed. When the Start button is pressed, the green LED will blink the same number of times. Pressing the Reset button will restart the system and clear the counter.
	Simple functions	Write a function void turnGreenOn() to turn the green LED on and call it in the main loop when the Start button is pressed.
	Functions with parameters	Create a function void blinkLED(int times) to blink the green LED a specified number of times and test it with different values.
Compound data types	Arrays and multidimensional Arrays in C++	Store the last 5 button presses in an array and use it to determine the LED behavior (e.g., blink green if the last press was Start, red if Emergency).
C++ Project	Header files and project structure in C++ and VSC	Complement it with the next topic. Create the Sensor.h and Sensor.cpp files to implement the sensors' code.

Outcomes:

- Understand C++ program structure and syntax
- Use variables, data types, and operators
- Implement control flow with conditions and loops
- Debug and fix simple code errors

Core Programming

This module focuses on programming the core functionalities of a mobile robot, specifically how to interact with sensors and actuators. Participants will learn how to write code that reads data from various sensors (e.g., IR, ultrasonic, line-followers) and controls outputs such as motors.

Topic	Subject	Task
Sensor Programming	Limit Switch	Implement the sensor's code and display the current measurement on Shuffleboard.
	Line Sensor	
	IR Sensor	

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	Ultrasonic Sensor	
	NavX Sensor	
Actuators Programming	DC Motor	Implement the actuators' code and interact with the robot's motors according to the sensors' measurements.
	Servo motor	

Outcomes:

- Explain how sensors collect data and actuators respond to commands.
- Write programs to read values from analog and digital sensors (e.g., IR, ultrasonic, light sensors).
- Use code to control DC motors and servos for movement (e.g., start, stop, speed, direction).
- Combine sensor input and actuator output in functional control loops (e.g., follow a line, avoid obstacles).

Fundamentals of Robotics

This module provides a foundational understanding of mobile robotics, focusing on the principles that govern the design, movement, and behavior of autonomous robots.

Topic	Subject	Task
Fundamentals	Encoder	Implement the sensor's code and display the current measurement on Shuffleboard.
	Wheels Velocity	Implement the code to obtain the speed of the motors in units of RPM, mm/s or m/s and display it on Shuffleboard.
Kinematics	Direct and Inverse Kinematics	Program a robot to move between points by calculating wheel velocities from desired motion coordinates. Program a robot to calculate its actual velocity using forward kinematics and display the measured speed in Shuffleboard.
	Kinematic Model	Implement a kinematic model to compute the robot's global position and orientation from local wheel velocities, enabling navigation in the workspace.
Odometry	Robot Position	Program the robot to calculate its real-time position by integrating measured wheel velocities, enabling accurate tracking of its movement in the workspace.
	Simple Control	Implement a control program that moves the robot along multiple paths by calculating trajectories from odometry and target coordinates.

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Outcomes:

- Understand the key components and characteristics of mobile robots.
- Explain how different wheel configurations (e.g., differential, omnidirectional) affect movement and control.
- Recognize the relation between electronics, mechanics, and programming in mobile robot functionality.
- Apply the kinematic model to compute wheel velocities and robot motion in real time.
- Use odometry data to track the robot's position and orientation within the workspace accurately.
- Integrate kinematic modeling and odometry to plan and execute precise robot trajectories between defined coordinates.

Image Processing

This module introduces participants to the fundamental concepts of image processing and how vision systems can be used in mobile robotics. It covers how digital images are formed, processed, and analyzed using software tools. Students will learn basic techniques such as filtering, color detection, edge detection, and object tracking.

Topic	Subject	Task
Python (Optional)	Python - Structure of a Program	Website Exercises and Examples
	Python - Variables, data types and constants	Website Exercises and Examples
	Python - Operators	Website Exercises and Examples
	Python - Basic Input/Output	Website Exercises and Examples
	Python - Conditional Structure	Website Exercises and Examples
	Python - Iteration Structures	Website Exercises and Examples
	Python - Functions	Website Exercises and Examples
Image Manipulation	Reading	Material Examples
	Visualizing	Material Examples
	Resizing	Material Examples
	Videos	Material Examples
Color Space	RGB/BGR	Colorful Shapes
	Conversion	Colorful Shapes
Filter	Kernels	DetectContours-Examples
	Blur	Blur -Examples
Color segmentation	InRange() and bitwise_and	DetectContours-Examples
Contour Detection	findContours()	Rice counting
	drawContours()	Cubes Detection
	boundingRect()	Tracking pacman
WPILIB Integration	Integrate Python Script w/ WPILIB	

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Outcomes:

- Explain how images are represented as pixel grids and how color models (RGB, grayscale) work.
- Apply transformations such as resizing, cropping, flipping, and rotating images.
- Capture and analyze frames from a live video stream using a webcam or onboard camera module.
- Identify common problems in image acquisition or processing and correct them.

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References

Studica WSR documentation page

https://docs.wsr.studica.com/en/latest/index.html
https://docs.dev.studica.com/en/latest/index.html

- Software's Install and Setup
- Control Station control
- VMX information
- Sensors and Actuators theory and Examples

Studica Training - WSA 2022 Mobile Robotics Training Robot Programming – C++

https://www.youtube.com/watch?v=H-acVbgdDnM

- Programming the robot
- Creating a new WorldSkills Project
- Driver Station and Dashboard Basics
- Creating Code to drive a basic robot

Studica Training - WSA 2022 Mobile Robotics Training Robot Build

https://www.youtube.com/watch?v=j7oVrAfu-Xc

- Building a simple four-wheel omni robot.

Studica Training – Building a basic Mobile Robotic - JAVA

https://www.youtube.com/playlist?list=PL6ErxHF11ImKbXdVJwJW0NQm5kb84eWKY

- Day 1 – Chassis, motor mounting, Structure, Wheels
- Day 2 – Elevator Mechanism, Lift, Servos
- Day 3 – Electrical, Sensors, Wiring, Control System
- Day 4 - Programming, Libraries, Software Tools, Driver Station, Autonomous Routines